

IT415 – APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES PRACTICAL EXAMINATION

Touchscreen Point of Sale (POS) Kiosk System

Student instructions and instructor checking guide

Scenario

A small campus food and merchandise outlet wants to introduce a touchscreen self-service Point of Sale Kiosk. Customers should select products by tapping large item buttons, review their order, choose a payment method, complete payment, and receive a digital receipt.

The current process requires a cashier to manually encode orders and compute totals. The outlet wants a simpler self-service interface similar to kiosks used in restaurants, payment centers, and retail stores.

Student task

Analyze the scenario and develop a functional Touchscreen POS Kiosk Application. Prioritize simple touch interaction, large clickable buttons, clear navigation, minimal typing, automatic computation, proper validation, and a clear transaction flow.

Technology choices

You may choose your own programming language, framework, database, UI library, development environment, application architecture, and AI development tool.

The application may be a web application, desktop application, Progressive Web App, mobile application, or local kiosk application. Your technology choice will not directly affect your grade as long as the required functionality works.

Required transaction flow

Step	Screen / action
1	Item Selection — tap products and adjust quantities.
2	Order / Payment Summary — review the complete order.
3	Payment Method — choose Cash, QR Payment, or Credit/Debit Card.
4	Payment Processing — validate and complete the selected payment.
5	Payment Successful — show confirmation and a transaction number.
6	Receipt — display the completed transaction.
7	New Transaction — reset and return to Item Selection.

Individual screens may be combined when appropriate, provided the required functionality and sequence remain clear.

1. Item Selection and Order Summary

Screen 1 — Item Selection

Display at least six products as large clickable or tappable item buttons/cards. Every product must show its name and price. Images, categories, descriptions, and remaining stock are optional. Users must not be required to type product names.

Example product button	Displayed price
Coffee	₱45.00
Sandwich	₱50.00
Soft Drink	₱35.00
Cookies	₱25.00
Bottled Water	₱20.00
Chocolate	₱25.00

These products and prices are examples only. Students may choose different products and prices.

Quantity controls

- Provide touch controls to increase and decrease quantity, such as – and + buttons.
- Allow the user to remove an item. Quantity must never become negative.
- If inventory is implemented, prevent selection beyond available stock.

Cart / current order

Clearly show the current order beside the products, at the bottom of the screen, or through a visible cart button. Display product name, unit price, quantity, and subtotal. Automatically update totals when items are added, changed, or removed.

Subtotal = Unit Price × Quantity

Total Amount = Sum of all item subtotals

Product	Quantity	Unit price	Subtotal
Coffee	2	₱45.00	₱90.00
Sandwich	1	₱50.00	₱50.00
Soft Drink	1	₱35.00	₱35.00
TOTAL			₱175.00

Provide a clearly visible Continue or Proceed to Payment button.

Screen 2 — Order / Payment Summary

Before payment, show all selected products, quantities, unit prices, subtotals, and the total amount. This screen must show the same order and total as Item Selection.

Provide Back to modify the order and Continue to Payment. Going back must preserve the selected items and quantities.

2. Payment Method and Processing

Screen 3 — Payment Method

Present at least three payment methods as large touchscreen buttons: Cash, QR Payment, and Credit/Debit Card. Actual payment gateway integration is not required; payments may be simulated, but the processing logic must work.

Cash payment

Show the total due, an amount-paid input, and a Pay Now button. Calculate the customer's change automatically.

Change = Amount Paid – Total Amount

Field	Example
Total amount	₱175.00
Amount paid	₱200.00
Change	₱25.00

Do not complete payment when the amount paid is lower than the total. Reject blank or invalid payment values and negative amounts. Keep the customer on the payment screen and show a clear message, such as “Insufficient payment. Please enter at least ₱175.00.”

Exact payment is valid and must produce ₱0.00 change.

QR payment

Show the amount to pay, a QR code or clearly identified QR placeholder, instructions to scan with a supported payment application, and a payment confirmation control.

You may use a generated QR code, static QR image, placeholder, or simulated payment confirmation. Real payment integration is not required. Once the simulated payment is confirmed, proceed to Payment Successful.

Example controls and message: “Scan the QR code using your supported payment application.” and “Confirm Payment”.

Credit / debit card payment

Show the amount due and an instruction such as “Please tap, insert, or swipe your card.” Provide a Process Payment control and a short processing state, such as “Processing payment...”.

After the simulated processing completes, display Payment Successful. Real card integration is not required.

Payment consistency

The amount shown during payment and on the receipt must match the confirmed order. For simulated QR and card payments, the amount paid equals the total and the change is ₱0.00.

3. Confirmation, Receipt, and Reset

Screen 4 — Payment Successful

After successful payment, show a clear confirmation containing the transaction amount, amount paid, payment method, transaction/reference number, and a View Receipt button.

Generate a reference for every completed transaction. It may be sequential, timestamp-based, or randomly generated. Unrelated completed transactions must not incorrectly share the same reference.

Screen 5 — Receipt

View Receipt must display a clean, readable digital receipt that accurately reflects the completed transaction. Include the transaction number, date, purchased items, quantities, unit prices or subtotals, total, payment method, amount paid, and change.

Sample digital receipt

CAMPUS STORE POS
Transaction No.: TXN-2026-00125
Date: October 6, 2026

Item	Quantity × unit price	Subtotal
Coffee	2 × ₱45.00	₱90.00
Sandwich	1 × ₱50.00	₱50.00
Soft Drink	1 × ₱35.00	₱35.00

Receipt field	Cash example	QR / card example
Total	₱175.00	₱175.00
Payment method	Cash	QR Payment / Credit/Debit Card
Amount paid	₱200.00	₱175.00
Change	₱25.00	₱0.00
Status	Payment Successful	Payment Successful

Exact receipt layout may vary. A screen-based digital receipt is acceptable; physical receipt printing is optional.

New Transaction

Provide a New Transaction button from the receipt. It must clear the cart, payment information, previous receipt, and transaction total, then return to Item Selection.

The previous customer’s order and payment details must not appear in the new transaction.

4. Interface and Required Functionality

Touchscreen interface requirements

Use large buttons, readable text, clear item cards, sufficient spacing, simple navigation, clear Back and Continue controls, minimal text input, and understandable icons or labels. Avoid tiny links, small controls, complex menus, excessive typing, and desktop-only interaction.

The interface should resemble a kiosk. Students are not required to reproduce a specific visual design.

Required functionality

1. Display at least six selectable products.
2. Allow products to be selected by clicking or tapping.
3. Allow quantity adjustment.
4. Allow removal of products.
5. Automatically calculate item subtotals.
6. Automatically calculate the transaction total.
7. Display an Order Summary.
8. Allow the user to return and modify the order.
9. Provide at least three payment method buttons.
10. Process Cash payment.
11. Validate insufficient Cash payment.
12. Calculate correct change.
13. Simulate QR payment.
14. Simulate Credit/Debit Card payment.
15. Display Payment Successful confirmation.
16. Generate and display a transaction number.
17. Display a digital receipt.
18. Allow the system to start a new transaction.
19. Properly reset the application after the transaction.
20. Provide meaningful user feedback.

User feedback

Clearly communicate results and errors, for example: “Product added”, “Invalid quantity”, “Insufficient payment”, and “Transaction completed successfully”. If inventory is implemented, show understandable feedback for insufficient stock.

5. Data, Enhancements, and Example

Data storage

You may use hard-coded product data, JSON files, Local Storage, SQLite, MySQL, PostgreSQL, Supabase, Firebase, MongoDB, or another appropriate approach. A database is not mandatory unless separately required by the instructor.

Be prepared to explain why your selected approach is appropriate for the application.

Optional enhancements

Product categories; product images; search; inventory deduction; transaction history; receipt printing; discounts, including Senior Citizen, PWD, or Student discounts; Cashier/Admin login; sales reports; dark/light theme; full-screen kiosk mode; product management; database persistence; actual QR generation.

Optional features must not replace the required functionality. Inventory deduction is optional in this touchscreen kiosk scenario; stock validation applies when inventory is implemented.

Expected complete transaction example

Selected item	Computation	Subtotal
Coffee × 2	2 × ₱45.00	₱90.00
Sandwich × 1	1 × ₱50.00	₱50.00
Soft Drink × 1	1 × ₱35.00	₱35.00
TOTAL		₱175.00

The customer reviews the ₱175.00 order summary, continues to Payment Method, selects Cash, and pays ₱200.00. The application calculates ₱25.00 change and displays Payment Successful.

View Receipt then shows a transaction reference, the three products with their quantities and subtotals, total ₱175.00, payment ₱200.00, change ₱25.00, and payment method Cash. New Transaction returns to an empty Item Selection screen.

Important note to students

Your application may use any visual style, but the transaction must naturally follow: Select Items → Review Order → Select Payment Method → Complete Payment → View Receipt → Start New Transaction.

The instructor will evaluate both the functional application and the actual development process demonstrated through AI-assisted development and Git/GitHub version control. Follow the separately issued requirements for AI documentation, commits, feature branches, pull requests, review, and merge.

A simple application that works correctly and can be explained clearly is preferable to a complicated application with incomplete features.

6. Instructor Tests — Selection and Order

Instructor checking guide. Use the sample prices below or calculate equivalent expected results using the student's product prices. Exact layouts and labels may vary.

Test	Action	Expected output / pass condition
1. Startup and touch selection	Open the application. Select products by tapping/clicking their cards.	Application runs. At least six products have visible names and prices. Controls are large enough for touch use; product names do not require typing.
2. Add multiple products	Add Coffee at ₱45 × 2, Sandwich at ₱50 × 1, and Soft Drink at ₱35 × 1.	Cart shows subtotals ₱90, ₱50, and ₱35. Total = ₱175.
3. Quantity controls	Increase Coffee from 2 to 3, then decrease back to 2.	Coffee subtotal becomes ₱135 and total becomes ₱220. Reducing back restores total ₱175. Quantity does not become negative.
4. Remove item	Remove Soft Drink from the ₱175 order.	Soft Drink disappears. Total automatically changes to ₱140.
5. Order Summary	Proceed to Order Summary.	Coffee = ₱90; Sandwich = ₱50; total = ₱140. Quantities, unit prices, and subtotals match Item Selection.
6. Back navigation	Select Back from Order Summary.	Return to Item Selection. Existing items and quantities are preserved and can be modified.
7. Payment Method	Continue from Order Summary to payment.	At least three large, tappable options appear: Cash, QR Payment, and Credit/Debit Card.

Validation observations

Check that the application prevents invalid quantities and gives clear feedback. If stock is implemented, check that the selected quantity cannot exceed stock.

An incomplete or invalid payment must not produce a successful transaction or receipt. When optional inventory deduction is present, rejected payments must not reduce stock.

7. Instructor Tests — Payment and Receipt

Test	Action	Expected output / pass condition
8. Insufficient Cash	For total ₱140, choose Cash and enter ₱100.	Reject payment with clear feedback. Remain in payment processing. Do not show a successful receipt.
9. Successful Cash	For the same ₱140 order, enter ₱200 and complete payment.	Change = ₱60. Proceed to Payment Successful. Exact payment must also be accepted with ₱0 change.
10. Confirmation	Inspect the successful payment screen.	Transaction amount, payment method, transaction/reference number, and View Receipt are visible.
11. Receipt	Select View Receipt for the Cash transaction.	Receipt contains reference, items, quantities, unit prices or subtotals, total ₱140, amount paid ₱200, change ₱60, and method Cash.
12. QR Payment	Start a new transaction, add items, and select QR Payment. Complete simulated confirmation.	Show amount, QR code/placeholder, and confirmation control. Payment succeeds. Receipt method = QR Payment.
13. Card Payment	Start another transaction and choose Credit/Debit Card. Process simulated payment.	Show tap/insert/swipe instruction and a processing state. Payment succeeds. Receipt method = Credit/Debit Card.
14. New Transaction	Select New Transaction from a receipt.	Return to Item Selection. Cart is empty; total = ₱0; payment and previous transaction details are cleared.
15. Transaction number	Complete two different transactions.	Each has a different transaction reference, such as TXN-001 and TXN-002. Format is up to the student.

For QR and card simulations, verify that the receipt amount paid equals the order total and change is ₱0.00. The receipt must represent the actual completed order and selected method.